

Differential Braking-Assisted Emergency Steering Function for Semi-trailer Trucks

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Abstract—Emergency steering maneuvers offer significant safety benefits by enabling the navigation of perilous traffic situations. Extensive research has been conducted to develop emergency maneuvering systems for passenger vehicles. However, performing evasive maneuvers with semi-trailer trucks presents unique challenges that limit their maneuvering capabilities, such as restrictions on the power steering system. This paper presents a unified control approach for automatic evasive maneuvering that integrates steering and differential braking. Differential braking, applied within conservative limits, is used to generate an additional yaw moment to complement the steering, thus enhancing the lateral dexterity of the vehicle. A jerk-optimal quintic polynomial path generation approach is adopted. For path tracking, a linear-quadratic controller (LQR) is employed. The proposed system is evaluated via simulations in MATLAB/Simulink. The simulation results show a significant improvement in evasive performance based on our assessment criteria, the Last Point to Brake (LPTB) and the Last Point to Steer (LPTS).

Index Terms—Emergency Steering Function, Collision Evasive Lateral Maneuver Systems, Evasive Steering Assist, Autonomous Driving, Advanced Driver Assistance Systems (ADAS), Semi-trailer Truck, Active Safety Systems, Vehicle Control, Differential Braking

I. INTRODUCTION

Evasive maneuvering-based collision avoidance systems have gained prominence in recent years as a strong complement to Autonomous Emergency Braking (AEB) systems, in scenarios where timely braking is no longer possible to prevent a collision. One prevalent issue with such systems is the inherent constraints on steering systems, which limit the scenarios in which such maneuvers can be executed. This has motivated the development of unified control of differential braking and steering to fully utilize the force capabilities of tires, and enhance maneuver dexterity. Differential braking is achieved through the actuation of brakes unilaterally to generate a yaw moment around the vertical axis of the vehicle.

Such a system was studied extensively by Cui et al. [1] in 2019. They proposed a 3-mode decoupled control structure for autonomous steering and differential braking control utilizing model predictive control (MPC), sliding mode control (SMC), and feedforward control. In contrast, Hajiloo et al. [2] developed a coupled control structure for steering and differential braking with MPC, incorporating actuation and tire force constraints. Another approach for differential braking control in the presence of steering system failures

was developed by Sung et al. [3]. They employed a constrained LQR where the optimal feedback gain matrix is obtained by solving the MPC problem offline [4], with a Kalman filter as a state estimator. Li et al. [5] utilized LQR as a trajectory generator from a given setpoint yaw rate and employed a PID controller for coupled steering and braking control. Hassanzadeh et al. [6] investigated such a system for heavy vehicles, accounting for rollover limits using 5th-order polynomial path generation.

While the following studies do not employ differential braking, they nonetheless provide valuable insight. Keller et al. [7], [8], in collaboration with TRW, focused on driver-initiated evasive steering systems within a driving simulator environment. Both studies utilized fifth-order polynomials for path generation, similar to [6]. In the first study [7], a nonlinear cascaded controller was employed, integrating feedforward, state feedback, and PD control elements. Furthermore, the second study [8] refined this approach using a cascaded controller architecture. In 2015, Gotte et al. [9], also collaborating with TRW, proposed a nonlinear-MPC approach, unifying path planning and motion control into an optimization problem. A similar study [10] from the same authors improved the real-time capabilities of the system by transitioning to a simpler vehicle model, enabling real-world applications. Hestermeyer et al. [11] presented the Ford Focus's lateral vehicle control features, which include evasive steering assist (ESA). It employs a proportional controller with path preview. Zhang et al. [12] from Aptiv, utilized an LQR controller for steering, and a fifth-order polynomial for path generation, which is similar to the aforementioned studies.

The aforementioned studies predominantly address passenger vehicles. A notable exception is [6], which investigates a rigid heavy vehicle. Our previous study [13] investigated the feasibility and applicability of evasive steering maneuvers for articulated semi-trailer trucks and concluded that the power steering system's restrictions render the maneuver ineffective. To address this challenge, in this study, a unified differential braking and steering control strategy is implemented to make evasive maneuvering feasible for this vehicle class, while the limitations of the power steering system and the vehicle are meticulously examined. Specifically, this work investigates the Type 1 Emergency Steering Function (ESF) as defined by ISO 23375:2023 [14]. This standard, along with UN Regulation No. 79 [15], categorizes ESF into Type 1 (fully automatic) and Type 2 (driver-initiated). The present work focuses on the Type 1 automatic activation

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approach.

The rest of the paper is organized as follows. Section II briefly describes the architecture and model of the system, controller design, and performance metrics. Section III presents the results of unified control of steering and differential braking compared to steering alone. Finally, concluding remarks are presented in Section IV.

II. BACKGROUND

A. Overall Architecture

The architecture of an evasive steering system designed to perform emergency maneuvers with the purpose of collision avoidance comprises several key subsystems. These subsystems are perception, path generation, and lateral control, in which the real-time decision-making and control are performed as a result of collective operation.

The perception subsystem is responsible for sensing the environment by interpreting measurement data collected from sensors such as: radar, lidar, and cameras. The environment includes objects, road delimiters, and other relevant features. Consequently, reliable perception is crucial as it is a key factor in determining the system's capability to perform safe evasive maneuvers.

Path generation is responsible for computing a collision-free and dynamically feasible path for the vehicle motion. This subsystem utilizes the environment information obtained by the perception subsystem to generate a path that enables safe and capable evasive maneuver.

The lateral control subsystem integrates both steering and differential braking efforts to accurately track a desired evasive path. This component also accounts for maintaining vehicle stability throughout the maneuver.

The vehicle subsystem, which encapsulates the vehicle and actuator dynamics, enables smooth execution of control signals. The vehicle's response is continuously monitored for the success of the evasive maneuver. Moreover, input constraints related to steering and differential braking are respected by the subsystem.

This unified architecture facilitates sensing, data processing, motion control, and accurate vehicle response, which are essential for various ADAS and autonomous driving applications. Each subsystem is essential for safe and effective collision avoidance across different driving scenarios, improving the overall performance and reliability of the entire system.

B. Truck-Trailer Linear Dynamic Model

Heavy commercial vehicles such as semi-trailer trucks can be effectively represented by single-track vehicle models. In this study, such a model is derived with the following assumptions: the longitudinal velocity is constant, the lateral tire force generated by steering is linear, and the articulation angle is small, which allows for the use of the small-angle approximation. Since the scope of this work is only interested in yaw-plane dynamics, these assumptions are considered justified. Fig. 1. illustrates the free body diagram of the fourth-order single track vehicle model. In this model, δ

denotes the steering angle. r_1 , and r_2 represent the corresponding yaw rates of the truck and semi-trailer, respectively. Similarly, ψ_1 and ψ_2 represent the corresponding yaw angles. γ indicates the articulation angle between the truck and semitrailer. a_1 and b_1 represent the distances to the center of gravity from the front axle and rear axle of the truck. a_2 and b_2 denote the distances to the center of gravity from the hitch point and the rear axle of the semi-trailer. Finally, H and c_1 denote the distances from the hitch point to the rear axle and the center of gravity of the truck.

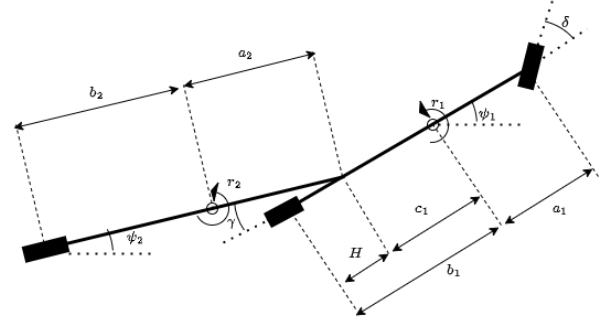


Fig. 1. Free body diagram of the single-track semi-trailer truck model. [13]

The single-track semi-trailer truck model utilized in this study is adopted from [16], [17], [18], [19]

The equations of motion along the body fixed-frame was developed in [17] as shown in (1)-(4).

$$m_1 a_{y1} = F_{yf1} + F_{yr1} - Y_h \quad (1)$$

$$I_{z1} \dot{r}_1 = a_1 F_{yf1} - b_1 F_{yr1} + c_1 Y_h \quad (2)$$

$$m_2 a_{y2} = F_{yt2} + Y_h \quad (3)$$

$$I_{z2} \dot{r}_2 = a_2 Y_h - b_2 F_{yt2} \quad (4)$$

Here, $r_2 = r_1 + \dot{\gamma}$. The lateral acceleration terms for the truck and semi-trailer are expressed as

$$a_{y1} = \dot{v}_{y1} + v_x r_1 \quad (5)$$

$$a_{y2} = \dot{v}_{y1} + v_x r_1 - c_1 \dot{r}_1 - a_2 (\dot{r}_1 + \ddot{\gamma}) \quad (6)$$

Tire force in semi-trailer equations is expressed as

$$F_{yf1} = C_{yf1} \alpha_{f1} \quad (7)$$

$$\alpha_{f1} = \delta - \frac{v_{y1} + a_1 r_1}{v_x} \quad (8)$$

$$F_{yr1} = C_{yr1} \alpha_{r1} \quad (9)$$

$$\alpha_{r1} = -\frac{v_{y1} - b_1 r_1}{v_x} \quad (10)$$

$$F_{yt2} = C_{yt2} \alpha_{t2} \quad (11)$$

$$\alpha_{t2} = -\frac{v_{y1} - (c_1 + a_2 + b_2) r_1 - (a_2 + b_2) \dot{\gamma}}{v_x} + \gamma \quad (12)$$

Keep in mind that the hitch force Y_h is not explicitly defined. To eliminate the Y_h from equations of motions, add (3) to (1). Multiply (1) by c_1 and add to (2). Then, multiply (3) by $-a_2$ and add to (4). Thus, the final form of equations of motion

are given as follows

$$m_1 a_{y1} + m_2 a_{y2} = F_{yf1} + F_{yr1} + F_{yt2} \quad (13)$$

$$I_{z1} \dot{r}_1 + c_1 m_1 a_{y1} = (c_1 + a_1) F_{yf1} + (c_1 - b_1) F_{yr1} \quad (14)$$

$$I_{z2} \dot{r}_2 - a_2 m_2 a_{y2} = -(a_2 + b_2) F_{yt2} \quad (15)$$

After substituting (8), (10), and (12) into (13), (14), and (15), and defining the state vector $x = [v_{y1} \ r_1 \ \dot{\gamma} \ \gamma]$ and the control vector $u = [\delta \ M_z]$, we can establish a state-space representation in the standard form $\dot{x} = Ax + Bu$ using (16)–(22). Here, A is the state matrix and B is the input matrix. M , D_m , G , C , F are coefficient matrices to revise (13)–(15) into a matrix-vector form.

$$M\dot{x} + D_m x = GCx + GFu \quad (16)$$

$$\dot{x} = \underbrace{M^{-1}(GC - D_m)}_A x + \underbrace{M^{-1}(GF)}_B u \quad (17)$$

where

$$M = \begin{bmatrix} m_1 + m_2 & -m_2(c_1 + a_2) & -m_2 a_2 & 0 \\ m_1 c_1 & I_{z1} & 0 & 0 \\ -m_2 a_2 & I_{z2} + m_2 a_2(c_1 + a_2) & I_{z2} + m_2 a_2^2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (18)$$

$$D_m = \begin{bmatrix} 0 & (m_1 + m_2)v_x & 0 & 0 \\ 0 & (m_1 c_1)v_x & 0 & 0 \\ 0 & -(m_2 a_2)v_x & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \quad (19)$$

$$G = \begin{bmatrix} 1 & 1 & 1 \\ a_1 + c_1 & c_1 - b_1 & 0 \\ 0 & 0 & -(b_2 + a_2) \\ 0 & 0 & 0 \end{bmatrix} \quad (20)$$

$$C = \begin{bmatrix} -\frac{C_{yf1}}{v_x} & -\frac{C_{yf1}a_1}{v_x} & 0 & 0 \\ -\frac{C_{yr1}}{v_x} & \frac{C_{yr1}b_1}{v_x} & 0 & 0 \\ -\frac{C_{yt2}}{v_x} & \frac{C_{yt2}(c_1 + b_2 + a_2)}{v_x} & \frac{C_{yt2}(a_2 + b_2)}{v_x} & C_{yt2} \end{bmatrix} \quad (21)$$

$$F = \begin{bmatrix} C_{yf1} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad (22)$$

At the same time, the dynamics of the steering system are expressed by the following differential equation.

$$\dot{\delta} = -\frac{1}{\tau}\delta + \frac{1}{\tau}\delta_{des} \quad (23)$$

where τ is the steering time constant and δ_{des} is the desired steering angle received by the power steering system. Furthermore, the yaw moment generated by differential braking can be obtained as:

$$M_z = \frac{1}{2}(F_{xf1}T_f + F_{xr1}T_r) \quad (24)$$

where F_{xf1} , F_{xr1} are front and rear brake forces, and T_f , T_r are the track widths of the front and rear axles, respectively.

C. Input Constraints

The industry-standard power steering system employed in this study is modeled, including its safety restrictions imposed on the system's external control interface. As the primary objective of this study is to make ESF maneuvers feasible despite steering limitations, these restrictions are rigorously applied to the model. The applied restrictions on the steering wheel angle and rate are used to enforce saturation limits on the control inputs as follows:

$$|\delta| \leq \delta_{max} \quad (25)$$

$$|\dot{\delta}| \leq \dot{\delta}_{max} \quad (26)$$

Likewise, the brake force utilized for differential braking must also be constrained by the lateral tire force generated by the steering. Given that the resultant force on the tire is limited by the normal load and the friction coefficient of the road, the allowable brake force for differential braking can be limited via the friction circle with a radius of $F_{max} = \mu F_z$ [20]. The available brake force can be expressed as follows:

$$|F_x| \leq \sqrt{(\mu F_z)^2 - F_y^2} \quad (27)$$

where the lateral tire forces can be obtained via equations (7) – (10), and substituting into (27) results in:

$$|F_{xf1}| \leq \sqrt{(\mu F_{zf1})^2 - (C_{f1}\alpha_{f1})^2} \quad (28a)$$

$$|F_{xr1}| \leq \sqrt{(\mu F_{zr1})^2 - (C_{r1}\alpha_{r1})^2} \quad (28b)$$

Depending on the load configuration of the vehicle, a conservative estimate for the rear and front axles $F_{x1f,max}$, $F_{x1r,max}$ is obtained for the maximum allowable brake forces. Therefore, the constraints can be written formally as:

$$|F_{xf1}| \leq F_{x1f,max} \quad (29a)$$

$$|F_{xr1}| \leq F_{x1r,max} \quad (29b)$$

D. Path Generation

The emphasis of this study is on the evasive performance of semi-trailer trucks from a vehicle dynamics perspective. Hence, the path generation approach is neither prioritized nor optimized in terms of computational efficiency.

In accordance with the regulation [15] and the standard [14], the maximum lateral displacement of the evasive action is limited for Type-1 systems, stipulating that the vehicle must not exceed the lane boundaries of its current lane. Therefore, an average lane clearance of 0.5 meters is assumed, based on standard road and vehicle dimensions [13]. Accordingly, this study employs evasive path precomputation with suitable boundary conditions that represent the desired evasive action.

The method implemented for the generation of the pre-computed path is adopted from the work presented in [21], which utilizes a jerk-optimal quintic polynomial [22]–[24] that yields a smooth and continuous lateral acceleration profile. As a result, undesirable transient load exerted on the steering system and tires is minimized, increasing simulation



Fig. 2. Autonomous Driving Development Truck based on Ford F-MAX. [25]

fidelity by avoiding operation in nonlinear regions. To isolate and focus on the dynamics of the maneuver, this study includes generating a quintic polynomial [21] for the desired evasive path for a straight road, avoiding additional load on vehicle subsystems resulting from curved roads.

$$y(x) = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 \quad (30)$$

Where x is the longitudinal distance traveled and $y(x)$ represents the lateral distance as a function of x and the polynomial coefficients $c = [c_0, c_1, \dots, c_5]$. To compute the path, appropriate boundary conditions are selected for the maneuver scenario, and the required operations are performed as described in [13].

For tracking the evasive path, the lateral error, heading error, and curvature at the nearest point on the reference path relative to the instantaneous truck position must be calculated. Accordingly, the precomputed global path is transformed into the body-fixed vehicle frame at each simulation step, and the lateral error e_y , the heading error e_θ , and the path curvature κ are computed as:

$$e_y = c_0 \quad (31)$$

$$e_\theta = \arctan(c_1) \quad (32)$$

$$\kappa = \frac{2c_2}{(1 + c_1^2)^{3/2}} \quad (33)$$

given that

$$y(0) = c_0 \text{ and } y'(0) = c_1 \text{ and } y''(0) = 2c_2 \quad (34)$$

E. Controller Design

The lateral controller for the proposed system is designed using the infinite-horizon continuous-time (LQR) [26] which minimizes the following quadratic cost function:

$$J = \int_0^\infty [(x - x_{ref})^T Q (x - x_{ref}) + u^T R u] dt \quad (35)$$

where Q and R are positive semi-definite and positive definite symmetric weighting matrices, respectively. The weighting matrices are designed according to Bryson's rule [27]. The optimal control law $u = -Kx - v$ that minimizes

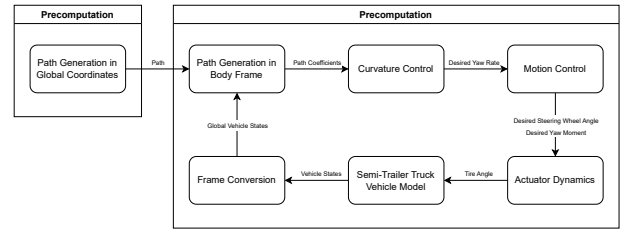


Fig. 3. Simulation architecture.

the cost function is obtained by solving the continuous-time algebraic Riccati equation as follows:

$$K = R^{-1} B^T P \quad (36a)$$

$$v = R^{-1} B^T (A^T - K^T B^T)^{-1} Q x_{ref} \quad (36b)$$

$$x_{ref} = [0 \quad r_{des} \quad 0 \quad 0]^T \quad (36c)$$

where P is the solution to the continuous-time ARE and the desired yaw rate is defined as $r_{des} = v_x(K_y e_y + K_\theta e_\theta + \kappa)$ heuristically. The gains K_y and K_θ are scheduled with respect to longitudinal velocity. Moreover, the pair (A, B) is controllable, fulfilling a fundamental requirement for this state-feedback approach.

F. Last Point to Brake and Last Point to Steer

The metrics LPTB and LPTS play a crucial role in the decision-making logic of an evasive steering system. LPTB is the distance required to avoid a collision or completely stop before an obstacle by applying braking. In the event of failure to apply braking at or before the last point can cause perilous situations. Likewise, LPTS is the latest point at which to initiate an evasive steering action to safely navigate around an obstacle. If this point is exceeded, the vehicle may either fail to generate adequate lateral motion or deviate excessively from the desired evasive path, resulting in undesirable outcomes. By comparing these key metrics, a decision is made as to whether to apply the brakes or initiate an evasive steering maneuver. The LPTB is calculated using the formula presented in [13]. Finally, to compute the LPTS values for the full speed range of the truck, the algorithm presented in our previous work [13] is employed.

III. SIMULATION RESULTS

In this section, the simulation architecture and its components are described. Subsequently, the effects of differential braking on evasive maneuvering are investigated. Finally, the evasive performance metrics LPTB and LPTS are evaluated to emphasize the feasibility of evasive steering compared to braking. The simulations are performed numerically in MATLAB/Simulink. The vehicle parameters mentioned in II-B are based on those of a Ford F-MAX MY 2018 shown in Fig. 2.

A. Simulation Architecture

The simulation architecture is designed to capture the essential dynamics and limitations of the truck-trailer system,

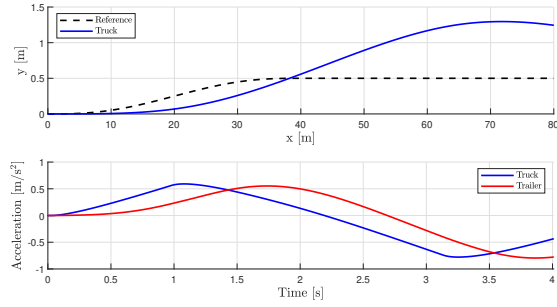


Fig. 4. Path tracking and lateral acceleration during steering-only controlled motion for 0.5 m lateral distance. [13]

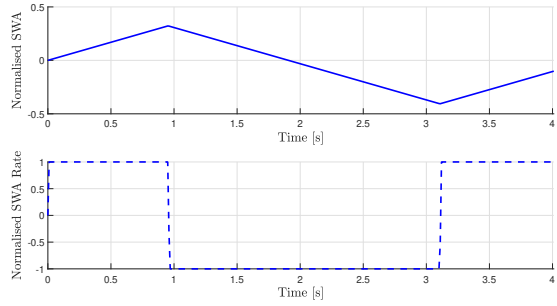


Fig. 5. Normalised steering wheel angle and rate during steering-only controlled motion for 0.5 m lateral distance. [13]

including the steering and differential braking actuators. A diagram of the architecture is depicted in Fig. 3. The constituent subsystems are as follows:

- **Path Generation in Global Coordinates:** This component generates the desired jerk-optimal path using a quintic polynomial for the vehicle to evade accordingly in global coordinates.
- **Path Generation in Body Frame:** The global path must be transformed into the vehicle body frame for control implementation. For perception systems that rely on camera-based perception, lane information is often provided as polynomials in the vehicle frame.
- **Curvature Control:** This component computes the desired yaw rate based on the lateral error, heading error, and curvature of the path. The lateral and heading errors are extracted from the planned path, which is expressed in polynomial form, according to equations (31) and (32), whereas road curvature is calculated using equation (33).
- **Motion Control:** This module integrates steering and differential braking using an LQR controller, which minimizes the path tracking error and overall control effort. LQR gains are recomputed for each simulation speed since the vehicle model is linearized at a constant longitudinal velocity.
- **Actuator Dynamics:** The aforementioned system limitations, including steering and differential braking dynamics, are represented in this component. Magnitude and rate-based saturation limits for steering wheel angle

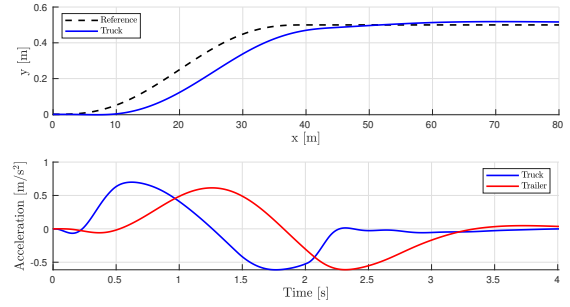


Fig. 6. Path tracking and lateral acceleration during steering and differential braking controlled motion for 0.5 m lateral distance.

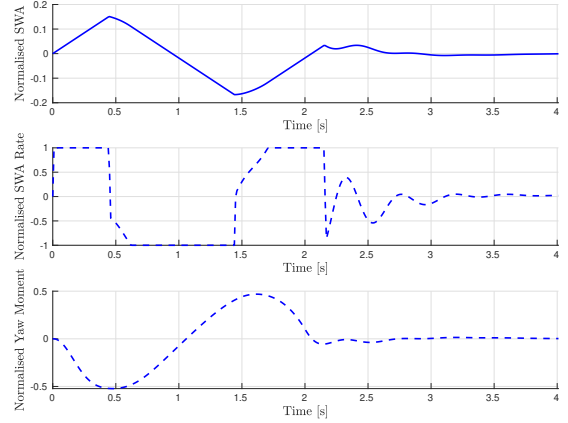


Fig. 7. Normalised steering wheel angle, rate, and yaw rate during steering and differential braking controlled motion for 0.5 m lateral distance.

and available brake forces are included in this component. Additionally, the response lag characteristics of the power steering system are represented by a first-order system.

- **Vehicle Dynamics:** This subsystem consists of an LPV state-space model simulating the truck-trailer's dynamic response, which includes lateral velocity, yaw rate, articulation angle, and articulation angle rate.
- **Frame Conversion:** The global position of the semi-trailer truck is calculated by this module to transform the reference path generated in the global frame to the body-fixed frame.

B. Effect of Differential Braking on Evasive Performance

This subsection is dedicated to the analysis of the effects of differential braking on the vehicle's evasive performance. The scenario defined for this purpose involves an evasive path with a lateral displacement target of 0.5 meters, a fully loaded semi-trailer truck traveling at a constant speed, and two different controller configurations. The first configuration is the steering-only motion control from our previous work [13]. The second configuration is the unified steering and differential braking motion control studied in this work. The initial conditions are $x(0) = (0, 0, 0, 0)$, and $v_x(0) = 20$.

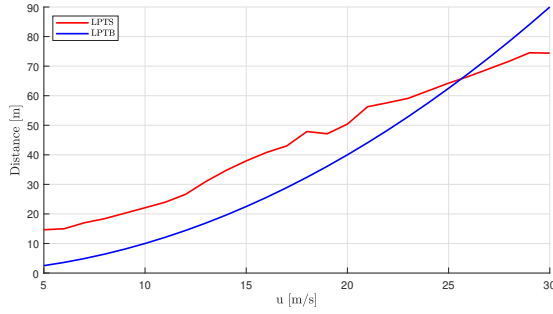


Fig. 8. LPTB and LPTS with steering only motion control for 0.5 m lateral distance. [13]

Figs. 4 and 6 show the reference paths for evasive maneuvers, the vehicle's tracking performance, and the lateral accelerations during the evasive maneuvers for both configurations. Figures 5 and 7 indicate the normalized steering wheel angles and rates similarly. In addition, the second configuration also includes the normalized yaw moment. Normalization is preferred as it facilitates a direct comparison of control inputs against their respective limits.

Fig. 4 shows that the first configuration could not achieve satisfactory results for the given reference path. Even though the lateral acceleration values are acceptable, the vehicle's response in lateral deviation exhibits an overshoot exceeding 100%. In addition, Fig. 5 indicates that the limit for the steering wheel angle rate is almost instantly reached, meaning that the maximum control effort satisfying the input constraints is insufficient for this maneuver [13]. On the other hand, Fig. 5 shows us that, with the addition of differential braking, the vehicle can track the same path with less than 5% maximum overshoot and a peak lateral acceleration of 0.75 m/s^2 , revealing that this maneuver is now possible. It is important to notice that Fig. 7 indicates that differential braking decreased the control effort on the steering system while simultaneously complying with the conservative brake force limits.

C. Evasive Performance Metrics

This section investigates evasive performance metrics by comparing LPTB and LPTS of a system with unified differential braking and steering control with the steering-only results from our previous study [13]. These two metrics are integral for the decision-making process during automated evasive maneuvers. By comparing these metrics, not only are the critical decision points for maneuver initiation evaluated, but the contribution of differential braking is quantified with results of the overall system performance in high-risk traffic scenarios.

Fig. 8 from our previous study [13] shows the LPTB and LPTS curves for the steering-only configuration. The curves intersect at approximately 26 m/s. This intersection indicates that initiating an evasive steering maneuver becomes more advantageous than braking only after the vehicle exceeds this longitudinal velocity. Therefore, we can deduce that the effectiveness of the steering-only ESF is limited to a

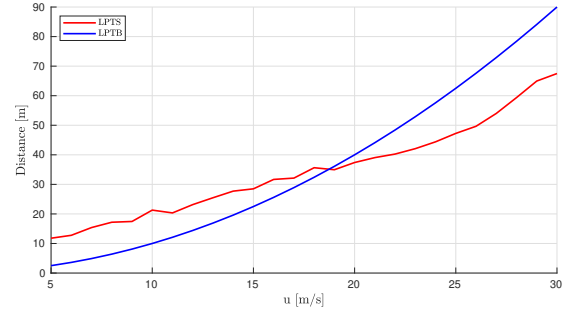


Fig. 9. LPTB and LPTS with steering and differential braking motion control for 0.5 m lateral distance.

narrow high-speed range, which is often above the speed limits of most highways worldwide for this vehicle category. This result indicates that an evasive steering maneuver is often nonviable and potentially hazardous within the typical operating speed range of semi-trailer trucks.

In contrast, Fig. 9 demonstrates the effect of differential braking where the intersection point of the curves is shifted from 26 m/s to approximately 18.5 m/s. Hence, the window of opportunity for ESF is considerably extended, reinforcing the effectiveness of an evasive steering maneuver over decelerating by only braking. Our proposed system not only achieved a 29% improvement in the required evasive maneuver starting distance but also mitigated the power steering system's limitations by bringing the ESF effectiveness range back to values nearly identical to the unrestricted steering case presented in [13]. Consequently, this system enables a safe and rewarding ESF solution that semi-trailer trucks may benefit from in a highway cruising scenario.

IV. CONCLUSION

This paper investigated the effects of differential braking combined with steering on Evasive Steering Function (ESF) performance. The proposed system employed a jerk-optimal polynomial-based path generation strategy in conjunction with a motion controller designed using a finite-horizon LQR, combining steering and differential braking control efforts to enhance evasive path tracking. The vehicle dynamics of a semi-trailer truck were modeled along with the input constraints that limit the vehicle's evasive capability. This work serves as a proof-of-concept to enable ESF for heavy commercial vehicles, which was deemed infeasible in a previous study.

The results show a significant improvement in the feasibility of ESF when employing a unified steering and differential braking strategy, which makes the ESF applicable to the typical cruising speeds of heavy commercial trucks. Furthermore, it is also shown that path tracking performance is greatly improved, and overshoots are minimized for demanding evasive paths. Consequently, the impacts of power steering system restrictions are mitigated, resulting in a more effective and feasible solution. The findings provide crucial insights into the future of evasive maneuvering for heavy commercial vehicles, thereby enhancing traffic safety.

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